QA TESTING PROJECT PORTFOLIO

Task 1 – Basic Selenium Automation

Task 2 – End-to-End Testing of eztrading.ae

TASK 1 – LEARN BASIC AUTOMATION USING SELENIUM

1. Project Information

Task Title: Learn Basic Automation Using Selenium

Automation Tool: Selenium WebDriver

Programming Language: Python

Browser: Google Chrome

Automation File: task1_search.py

Testing Type: Basic Web Automation

2. Objective

The objective of this task is to learn the basics of web automation using Selenium and Python. The automation script opens a web page, performs a search, and verifies that the expected search result is displayed successfully.

3. Task Scope

- Open a web page automatically using Selenium.
- Locate the required web element.
- Enter search information.
- Perform the search action.
- Wait for the expected result.
- Verify the result using Selenium.
- Display a PASS or FAIL result.

4. Tools and Technologies

- Python 3.11.0
- Selenium WebDriver
- Google Chrome
- Chrome WebDriver

5. Automation Flow

1. Start the Python script.
2. Launch Google Chrome using Selenium.
3. Open the target web page.
4. Locate the search field.

5. Enter the search term.
6. Perform the search.
7. Wait for the search result.
8. Verify that the expected result is displayed.
9. Print the test result.
10. Close the browser.

6. Test Case

Test Case ID	Test Case / Objective	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-001	Verify Web Search Using Selenium	Python, Selenium, and Google Chrome are installed.	1. Run the Selenium Python script. 2. Open the target web page. 3. Locate the search field. 4. Enter the search term. 5. Perform the search. 6. Verify the displayed result.	The web page should open successfully, the search should be performed, and the expected result should be displayed.	The webpage opened successfully, the search was performed, and the expected result was verified successfully.	PASS

7. Test Execution

Execution Method: Automated execution using Selenium and Python.

Execution Result: PASS

Console Result: SEARCH TEST PASSED

The automation script successfully opened the web page, performed the search, verified the result, and returned a successful PASS result.

8. Automation Script

Automation File: task1_search.py

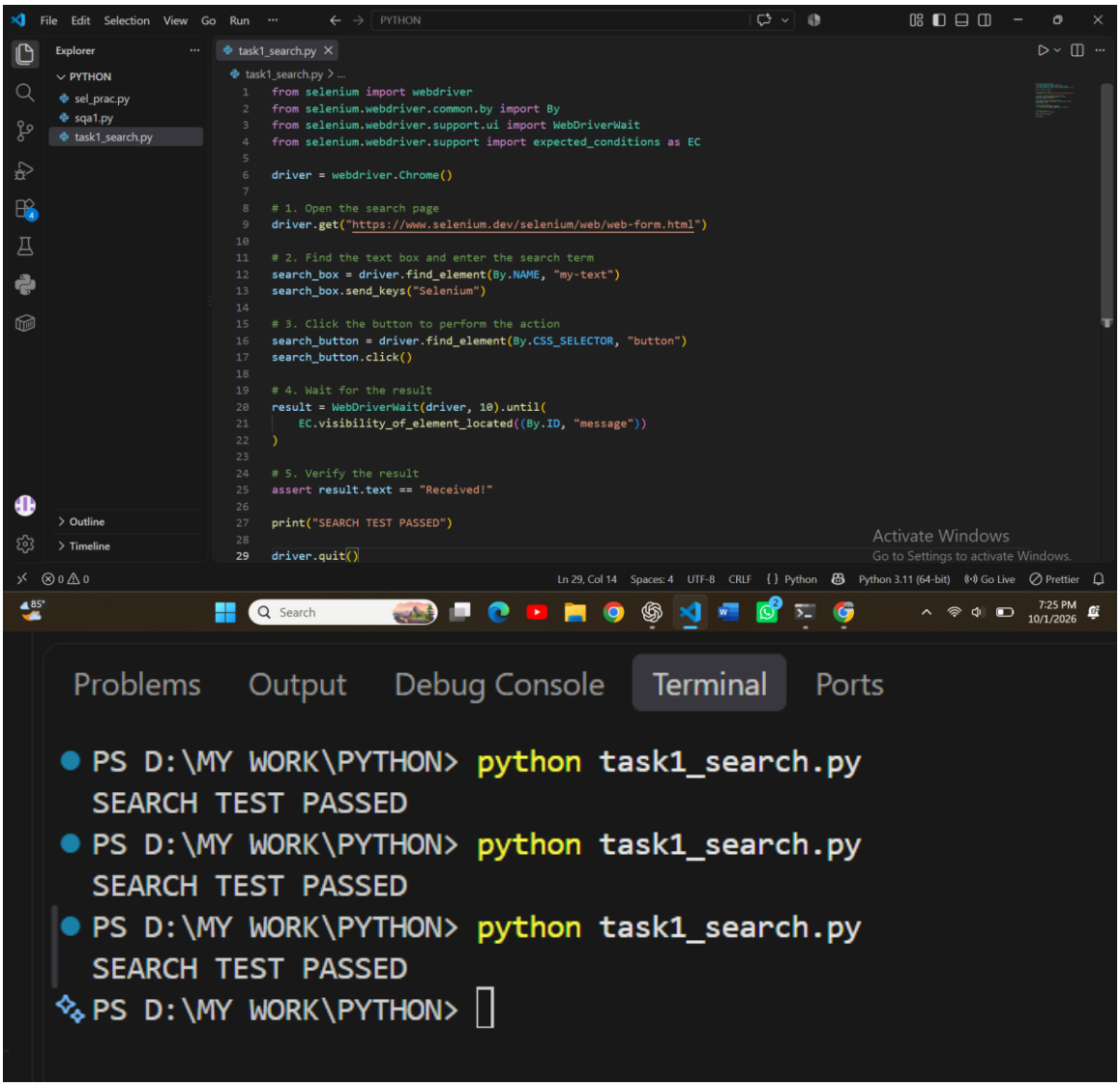
The script uses Selenium WebDriver to interact with the browser, locate the required elements, perform the search, wait for the expected result, and verify the outcome.

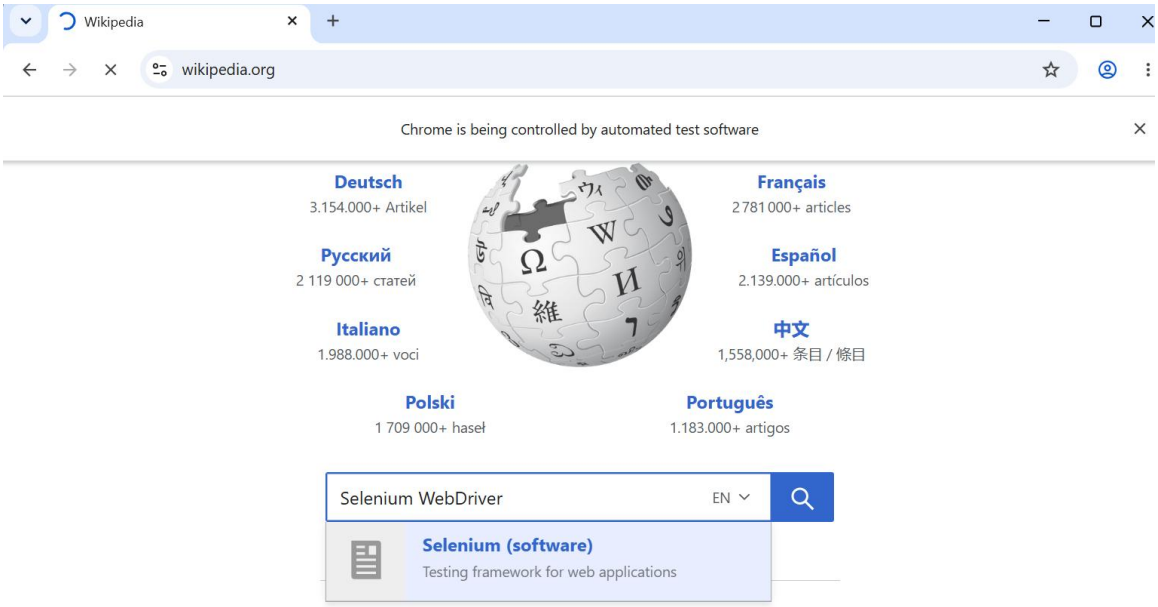
9. Verification

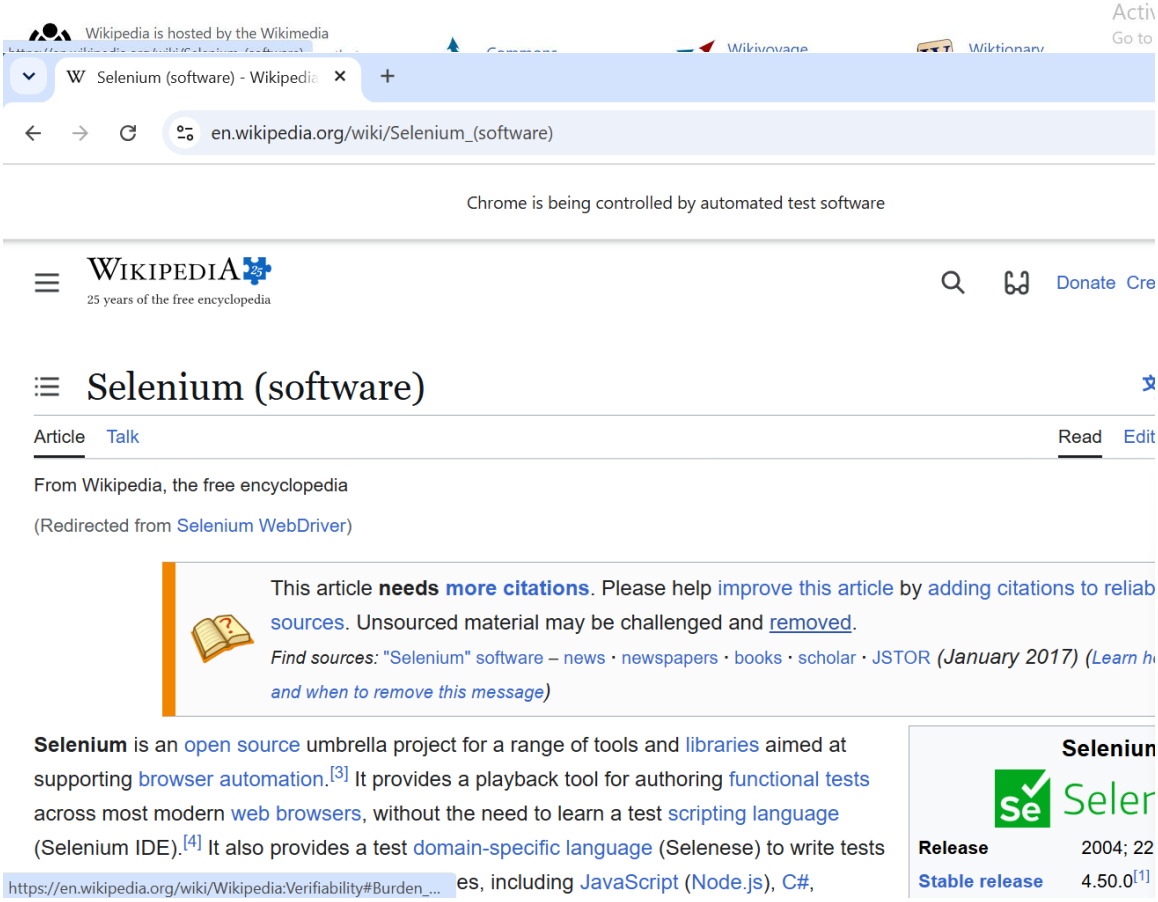
- Browser launched successfully.
- Target webpage opened successfully.
- Search field was located successfully.
- Search action was performed successfully.
- Expected result was found.
- Automation test returned a PASS result.
- Browser was closed after execution.

10. Evidence

Evidence for this task can include:




Wikipedia
wikipedia.org
Chrome is being controlled by automated test software
Deutsch
3.154.000+ Artikel
Français
2 781 000+ articles
Русский
2 119 000+ статей
Español
2.139.000+ artículos
Italiano
1.988.000+ voci
中文
1,558,000+ 条目 / 條目
Polski
1 709 000+ haseł
Português
1.183.000+ artigos
Selenium WebDriver
EN
Selenium (software)
Testing framework for web applications


Wikipedia is hosted by the Wikimedia
Selenium (software) - Wikipedia
en.wikipedia.org/wiki/Selenium_(software)
Chrome is being controlled by automated test software
WIKIPEDIA
25 years of the free encyclopedia
Donate
Selenium (software)
Article
Talk
Read
Edit
From Wikipedia, the free encyclopedia
(Redirected from Selenium WebDriver)
This article needs more citations. Please help improve this article by adding citations to reliab
sources. Unsourced material may be challenged and removed.
Find sources: "Selenium" software – news · newspapers · books · scholar · JSTOR (January 2017) (Learn h
and when to remove this message)
Selenium is an open source umbrella project for a range of tools and libraries aimed at supporting browser automation.[3] It provides a playback tool for authoring functional tests across most modern web browsers, without the need to learn a test scripting language (Selenium IDE).[4] It also provides a test domain-specific language (Selenese) to write tests
https://en.wikipedia.org/wiki/Wikipedia:Verifiability#Burden_...
es, including JavaScript (Node.js), C#,
Release
2004; 22
Stable release
4.50.0[1]

11. Result Summary

Item	Result
Webpage Opening	PASS
Search Action	PASS
Search Result Verification	PASS
Selenium Automation	PASS
Overall Task Result	PASS

12. Conclusion

The task was completed successfully using Selenium WebDriver with Python. The automation script successfully opened the web page, performed a search, verified the expected result, and displayed a successful PASS result. This task provided practical experience with basic Selenium concepts, browser automation, element interaction, result verification, and automated test execution.

TASK 2 – END-TO-END TESTING OF A WEB APPLICATION

1. Project Information

Project Name: End-to-End Testing of eztrading.ae

Application Under Test: eztrading.ae

Application Type: E-Commerce Web Application

Website: https://eztrading.ae/

Testing Approach: Manual Testing + Selenium Automation

Automation Tool: Selenium with Python

2. Objective

The objective of this project is to perform end-to-end testing of the eztrading.ae e-commerce website and verify that its major functionalities work correctly. The testing covers the user journey from browsing and searching for products to adding products to the cart and proceeding through the checkout process.

3. Scope of Testing

- Homepage
- Product search
- Product categories
- Product details
- Add to cart
- Cart functionality
- Product quantity
- Remove product from cart
- Login validation
- Checkout navigation
- Order summary
- Negative testing
- Selenium automation

4. Testing Types

- Functional Testing
- UI Testing
- End-to-End Testing

- Negative Testing
- Manual Testing
- Basic Automation Testing

5. Test Scenarios

Homepage

- Verify that the homepage loads successfully.
- Verify navigation and website elements.

Product Search

- Verify search for an existing product.
- Verify search for a non-existing product.
- Verify partial product search.
- Verify empty search.

Product Categories

- Verify category navigation.
- Verify products are displayed under the selected category.

Product Details

- Verify product name.
- Verify product price.
- Verify product image.
- Verify product description.
- Verify product availability.
- Verify Add to Cart functionality.

Shopping Cart

- Verify product can be added to cart.
- Verify product appears in cart.
- Verify product quantity can be updated.
- Verify product can be removed.
- Verify cart total.

User Account

- Verify login functionality.
- Verify invalid login validation.

Checkout

- Verify checkout navigation.
- Verify customer information.
- Verify address information.
- Verify order summary.
- Verify payment flow.

6. Test Cases

Test Case ID	Test Case / Objective	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-001	Verify Homepage Loading	Internet connection available	Open eztrading.ae and wait for the homepage to load	Homepage should load successfully without major errors	Homepage loaded successfully without major errors	PASS
TC-002	Verify Product Search	Website is open	Search for an existing product	Relevant product/results should be displayed	Relevant product/results were displayed	PASS
TC-003	Verify Product Category	Website is open	Select any product category	Selected category should open and display relevant products	Selected category opened and relevant products were displayed	PASS
TC-004	Verify Product Details	Website is open	Open a product and check name, price, image and description	Product information should be displayed correctly	Product information was displayed correctly	PASS
TC-005	Add Product to Cart	Product is available	Open a product → Click Add to Cart → Open cart	Selected product should appear in the cart with correct information and price	Product was added to the cart with correct information and price	PASS
TC-006	Update Product Quantity	Product is already in cart	Open cart → Increase product quantity → Check quantity and total	Quantity and total price should update correctly	Quantity and total price updated correctly	PASS
TC-007	Remove Product from Cart	Product is already in cart	Open cart → Click Remove/Delete	Selected product should be removed from the cart	Product was removed from the cart successfully	PASS

TC-008	Verify Login Validation	Login page is available	Enter invalid credentials → Click Login/Sign In	Invalid credentials should be rejected with an appropriate message	Invalid credentials were rejected and appropriate validation was displayed	PASS
TC-009	Verify Checkout Navigation	Product is in cart	Open cart → Click Checkout/Proceed	User should be directed to the checkout process	User was directed to the checkout process successfully	PASS
TC-010	Verify Order Summary	Product is in cart and checkout is reached	Review product, quantity, price and total	Order summary should display correct product, quantity, price and total	Order summary displayed the correct information	PASS
TC-011	Verify Search with Non-Existing Product	Website is open	Enter a random/non-existing product name → Click Search	Website should clearly indicate that no matching product was found	Other products were displayed instead of a 'No Products Found' message	Observation

7. Test Execution Summary

Total Test Cases: 11

Passed: 10

Failed: 0

Observations: 1

Bugs Identified: 1

Bugs Logged in Jira: 1

The test cases were executed manually on eztrading.ae. The major tested functionalities worked as expected.

8. Bug Report

Bug ID	Module	Bug Title	Steps to Reproduce	Expected Result	Actual Result	Severity	Priority	Status
BUG-	Product	Non-	1. Open	Website	Other	Medium	Medium	Open

001	Search	existing product search displays other products	eztrading.ae 2. Enter a random/non-existing product name 3. Click Search	should display a clear 'No Products Found' message or indicate that no matching product exists	products are displayed instead of a 'No Products Found' message			

9. Jira Bug Tracking

Bug ID: BUG-001

Issue Type: Bug

Title: Non-existing product search displays other products

Priority: Medium

Status: Open

Jira Action: Bug successfully created and logged in Jira.

10. Automation Testing

Automated Test Case: TC-002

Test Case: Verify Product Search

Automation Tool: Selenium

Programming Language: Python

Automation File: task2_search.py

Automation Flow

11. Open eztrading.ae.
12. Locate the search box.
13. Enter a product name.
14. Perform the search.
15. Verify that search results are displayed.
16. Display the test result.

Automation Result: TEST PASSED: Product search is working.

The Selenium automation script successfully opened the website, searched for a product, verified the search results, and returned a PASS result.

11. Test Plan

Test Plan Objective

The purpose of this test plan is to define the testing approach, scope, objectives, and activities used to evaluate the eztrading.ae e-commerce website.

Testing Scope

Testing covered the major shopping and user-related functionalities of the website, including product search, categories, product details, cart, login validation, checkout navigation, and order summary.

Testing Approach

Manual testing was used to test the major functionalities and identify possible issues. Selenium with Python was used to automate the product search test case.

Testing Environment

Application: eztrading.ae

Browser: Google Chrome

Testing Platform: Desktop

Automation: Selenium + Python

Entry Criteria

Testing was started after the website was accessible and the required testing environment was available.

Exit Criteria

- Major test cases were executed.
- Test results were recorded.
- Identified issue was documented.
- Bug was logged in Jira.
- Automation test was successfully executed.
- Final summary was prepared.

12. Overall Test Results

Metric	Result
Total Test Cases	11
Passed	10

Failed	0
Observations	1
Bugs Logged	1
Automated Tests	1

13. Conclusion

End-to-end testing was performed on the major functionalities of the eztrading.ae e-commerce website. The testing covered homepage loading, product search, product categories, product details, cart functionality, login validation, checkout navigation, and order summary.

Most of the tested functionalities worked as expected. One observation was identified in the handling of non-existing product searches, where other products were displayed instead of a clear 'No Products Found' message. This issue was documented and logged in Jira for further review.

In addition, the product search functionality was successfully automated using Selenium and Python.

Overall, the testing provided practical experience in manual testing, test case design, test execution, bug reporting, Jira bug tracking, and basic Selenium automation.